
EECS 16A Designing Information Devices and Systems I

Discussion 2B

1. Gaussian Elimination

Gaussian Elimination is a systematic procedure that a computer could follow for solving a large system of linear equations simultaneously. The augmented matrix is in the form:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right] \quad (1)$$

In Gaussian Elimination, we perform three operations on the system of equations. None of these three operations change the solution set of the system of equations. These operations are:

- (a) Multiplying an equation by a *nonzero* scalar constant.
- (b) Adding a scalar constant multiple of one equation to another.
- (c) Swapping two equations.

Gaussian Elimination Algorithm for augmented matrix (1):

Step 1: For $i = 1, 2, \dots, m$

- (i) If necessary, swap row i with a row below it, so that the leading entry in row i is as far left as possible.
- (ii) Rescale row i so that its leading entry is equal to 1.
- (iii) For rows $j = i + 1, i + 2, \dots, m$, add to row j a scalar multiple of row i , so that the leading entry of row i has all zeros below it.

Step 2: For $i = m, m - 1, \dots, 1$

- (i) Add to each row $j = 1, 2, \dots, i - 1$ a scalar multiple of row i so that the leading entry of row i has all zeros above it.

For the following systems, use Gaussian Elimination to solve the problem. Does a solution exist? Is it unique? If there are an infinite number of solutions, give the solution in the parametric form.

(a)

$$\begin{bmatrix} 2 & 0 & 4 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \\ 3 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 4 & 2 \\ 1 & 2 & 8 \\ 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$$

(c)

$$\begin{bmatrix} 2 & 2 & 3 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$$

2. Finding The Bright Cave

Nara the one-handed druid and Kody the one-handed ranger find themselves in dire straits. Before them is a cliff with four cave entrances arranged in a square: two upper caves and two lower caves. Each entrance emits a certain amount of light, and the two wish to find exactly the amount of light coming from each cave. Here's the catch: after contracting a particularly potent strain of ghoul fever, our intrepid heroes are only able to see the total intensity of light before them (so their eyes operate like a single-pixel camera). Kody and Nara are capable adventurers, but they don't know any linear algebra – and they need your help.

Kody proposes an imaging strategy where he uses his hand to completely block the light from two caves at a time. He is able to take measurements using the following four masks (black means the light is blocked from that cave):

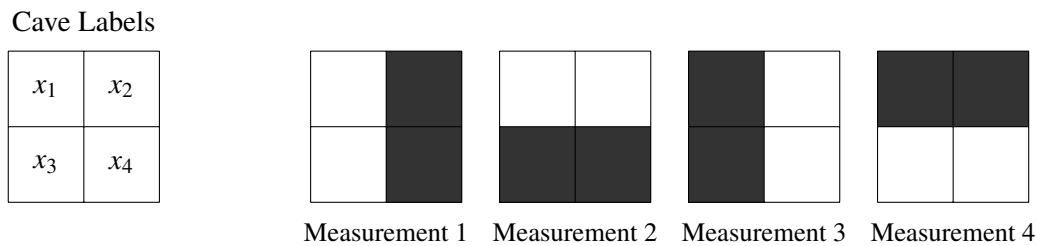


Figure 1: Four image masks.

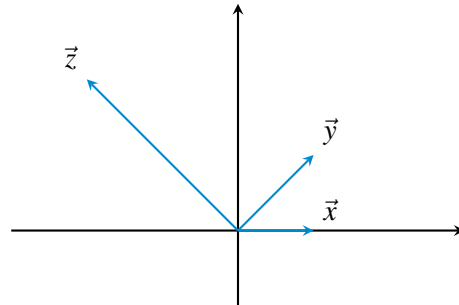
- (a) Let $\vec{x}$ be the four-element vector that represents the magnitude of light emanating from the four cave entrances. Write a matrix $\mathbf{K}$ that performs the masking process in Figure 1 on the vector $\vec{x}$, such that $\mathbf{K}\vec{x}$ is the result of the four measurements.
- (b) Does Kody's set of masks give us a unique solution for all four caves' light intensities? Why or why not?
- (c) Nara, in her infinite wisdom, places her one hand diagonally across the entrances, covering two of the cave entrances. However, her hand is not wide enough, letting in 50% of the light from the caves covered and 100% of the light from the caves not covered. The following diagram shows the percentage of light let through from each cave:

50%	100%
100%	50%

Does this additional measurement give them enough information to solve the problem? Why or why not?

3. Visualizing Span

We are given a point $\vec{z}$ that we want to get to, but we can only move in two directions: $\vec{x}$ and $\vec{y}$. We know that to get to $\vec{z}$, we can travel along $\vec{x}$ for some amount α , then change direction, and travel along $\vec{y}$ for some amount β . We want to find these two scalars α and β , such that we reach point $\vec{z}$. That is, $\alpha\vec{x} + \beta\vec{y} = \vec{z}$.



- (a) First, consider the case where $\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\vec{y} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, and $\vec{z} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$. Draw these vectors on a sheet of paper.
- (b) We want to find the two scalars α and β , such that by moving α along $\vec{x}$ and β along $\vec{y}$ so that we can reach $\vec{z}$. Write a system of equations to find α and β in matrix form.
- (c) Solve for α, β .